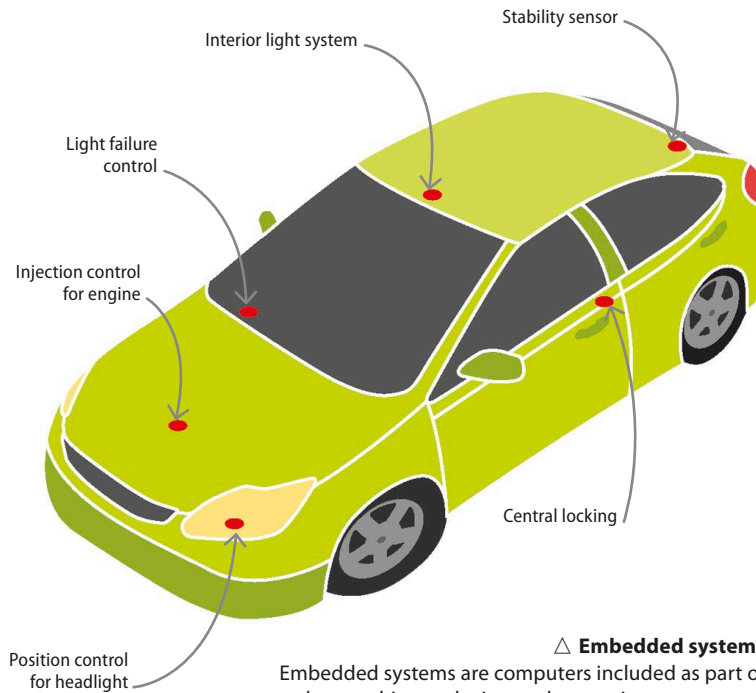


High- and low-level languages

The term “programming languages” is usually used for high-level languages. These allow programmers to use a language closer to human language. Low-level languages work with internal hardware like registers and memory, and are tied to a specific type of computer. Programs written in a high-level language can be run on any computer with the relevant compiler or interpreter.



△ Embedded systems

Embedded systems are computers included as part of another machine or device, such as a microwave, or car.

They are often programmed in C as, despite being a high-level language, it allows programmers a great deal of control over how the code runs at the level of registers.

TOP TECH

The Mars Rover

One of the most famous devices to feature an embedded system is the Mars Rover, *Curiosity*. The self-propelling robot is programmed to explore Mars and send back data. Code for the rover is written mainly in C and has been very thoroughly tested to try to ensure the rover doesn't accidentally drive into a rock and damage itself.



Special purpose

In the initial stages of computing, programs were written in binary code, or assembly language. Since then, programming languages have been developed as tools to meet a need or fulfil a purpose. Examples include languages that allow mathematicians and scientists to include formulas, languages used to teach people how to program, or languages that could be used to develop artificial intelligence.



FORTRAN

△ Scientific computing

Fortran was designed to allow scientists to write programs that included mathematical formulas. Its name is short for “Formula Translation”.



COBOL

△ Down to business

Short for common business-oriented language, COBOL was developed to make it easier and cheaper for companies to write business-related software.



Scratch

△ Code for kids

Scratch was created as a language that would make learning to code easy and fun for children aged between eight and 16.



Lisp

△ Thinking machines

Lisp was based on a mathematical definition of programming languages and soon became popular with researchers studying artificial intelligence.